

Technology #135

Integrate acoustic emission sensors along the infusion tubing that analyze pressure wave signatures characteristic of occlusion events

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

Generated: 2026-08-30

Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Integrate acoustic emission sensors along the infusion tubing that analyze pressure wave signatures characteristic of occlusion events

WHY DOES IT MATTER?

Documented recurring problem 'downstream occlusion detection failure' for infusion pumps: 100 MAUDE adverse-event records retrieved live at benchmark time (2007–2032). Event narratives: 'custody in retrieval log'. Literature context: 10 EuropePMC records on infusion-pump occlusion alarms. MAUDE LIMITS STAMPED INLINE: incidence UNKNOWN, causation UNVERIFIED, voluntary reports (Art. XXI.5).

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 13 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in fluidics_hydraulic. 10-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 3 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 10-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Expose the system to various occlusion scenarios (partial, complete, intermittent) with different fluid viscosities and flow rates, measu.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: bench flow-loop characterization across operating range. Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects; transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb; $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement; FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.; FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.; FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.; Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: Q (m³/s), Re (-), Q (m³/s) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r , m] value_status=UNKNOWN; lumen length [r , m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP , Pa] value_status=UNKNOWN; flow rate range [Q , m³/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

3 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: Industrial fluid dynamics monitoring system using pressure wave pattern recognition

POTENTIAL VALUE DRIVER: Addresses: Documented recurring problem 'downstream occlusion detection failure' for infusion pumps: 100 MAUDE adverse-event records retrieved live at benchmark time (2007–2032). Event narratives: 'custody in retrieval log'. Literature context: 10 EuropePMC records on infusion-pump occlusion alarms. MAUDE LIMITS STAMPED INLINE: incidence UNKNOWN, causation UNVERIFIED, voluntary reports (Art. XXI.5).

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: fluidics_hydraulic

Disciplines: fluid mechanics, mechanical engineering, biomedical engineering, materials science

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Integrate acoustic emission sensors along the infusion tubing that analyze pressure wave signatures characteristic of occlusion events

Subsystems:

?: flow path / lumen architecture - ?

?: pressure regulation element - ?

?: sensing / feedback element (if closed-loop) - ?

?: termination interfaces (patient / reservoir) - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Integrate acoustic emission sensors along the infusion tubing that analyze pressure wave signatures characteristic of occlusion events

Key Physics: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * dP) / (8 * \mu * L)$.

Applicability to this invention: APPLICABLE; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: dP, μ .

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention: APPLICABLE; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: μ .

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * dP / \rho}$. Applicability to this invention: APPLICABLE; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. Engaged variables: dP.

Assumptions:

- CSF approximated as Newtonian (valid at shunt flow rates)
- no wall compliance effects
- transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb
- $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement
- FLUID-001 (Hagen-Poiseuille (laminar pipe flow)) is applicable only under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow.
- FLUID-002 (Reynolds number) is applicable only under: circular section; single-phase flow.
- FLUID-003 (Orifice flow (sharp-edged restriction)) is applicable only under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured.
- Model input variables requiring sourced values before any numeric result: r (m), dP (Pa), μ (Pa.s), L (m), ρ (kg/m³), v (m/s), D (m), μ (Pa.s), C_d (-), A (m²), dP (Pa), ρ (kg/m³) — all UNKNOWN (no sourced value exists; Art. VIII).

- Model output variables produced by the governing model: Q (m^3/s), Re (-), Q (m^3/s) — UNKNOWN until the inputs are sourced and the model is evaluated.
- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): lumen inner diameter [r, m] value_status=UNKNOWN; lumen length [r, m] value_status=UNKNOWN; surface roughness [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; operating pressure head [dP, Pa] value_status=UNKNOWN; flow rate range [Q, m^3/s] value_status=UNKNOWN; viscosity of flowed medium [μ , Pa.s] value_status=UNKNOWN; occlusion growth tolerance [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'downstream occlusion detection failure', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'proximal occlusion by tissue ingrowth', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'lumen collapse under external pressure (kink)', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'encrustation / mineral deposition', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'regime transition to turbulence altering calibration', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'check-valve seat wear opening reverse leakage', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Occlusion detection must trigger before clinically harmful pressure/volume	SOURCE: FATE
DI-002	Mechanism driver (proposed)	Industrial fluid dynamics monitoring system using pressure wave pattern recognition	MODELED
DI-003	Expected effect target	Enhanced detection of partial and complete occlusions by recognizing distinctive	MODELED
DI-004	Domain design input pattern: required flow applicability	Variable pressure [pressure]	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: medium viscosity applicability	Variable viscosity [viscosity]	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: anatomical applicability	NOT_ESTABLISHED (candidate pattern; confirm against anatomy)	ENGINEERING_PROPOSED
DI-007	Domain design input pattern: device lifetime applicability	NOT_ESTABLISHED (candidate pattern; confirm against device)	ENGINEERING_PROPOSED
DI-008	Regulatory design input: compliance candidate standards	ISO 13485 (QMS) and FDA 21 CFR 820 (QMS) and EN ISO 13485 (QMS) and FDA 21 CFR 820 (QMS)	ENGINEERING_PROPOSED
DI-009	Custodied observation: Selected large Arterial Models of Vascular Arteries	content_hash 8ed456da0d9609d8c35e5a261ee31731beafa73c5ddfb3c	
DI-010	Custodied observation: Troubleshooting pumps in patients	content_hash 475700279218675807e5a70763ec954e069f046146c9a5	
DI-011	Custodied observation: Processes of the pump	content_hash 165b59001250d85c06e40b7e81b2710394a6ec6ca3fe	
DI-012	Custodied observation: Closed-loop pump system	content_hash 606680212107e79a77e22ab2193ce54d32d668db	
DI-013	Custodied observation: Hepatic Arterial Models of Vascular Arteries	content_hash 42658248e0a45e43e0d450b10f55469f0d6f05f8b5374f749a3c	

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	flow path / lumen architecture realizing Integrate acoustic emission sensors along the flow path	CONCEPTUAL	details of sensor placement and data analysis
DO-002	pressure regulation element realizing Integrate acoustic emission sensors along the flow path	CONCEPTUAL	details of sensor placement and data analysis
DO-003	sensing / feedback element (if closed-loop) realizing Integrate acoustic emission sensors along the flow path	CONCEPTUAL	details of sensor placement and data analysis
DO-004	termination interfaces (patient / reservoir) realizing Integrate acoustic emission sensors along the flow path	CONCEPTUAL	details of sensor placement and data analysis
DO-005	flow path / lumen architecture interfaces with pressure regulation element: signal transfer	CONCEPTUAL	details of sensor placement and data analysis
DO-006	pressure regulation element interfaces with sensing / feedback element (if closed-loop): signal transfer	CONCEPTUAL	details of sensor placement and data analysis
DO-007	sensing / feedback element (if closed-loop) interfaces with termination interfaces: signal transfer	CONCEPTUAL	details of sensor placement and data analysis
DO-008	Device boundaries: patient/fluid inlet and outlet termination control/data processing	CONCEPTUAL	details of sensor placement and data analysis
DO-009	Flow/force/signal path geometry realizing the mechanism	UNKNOWN	details of sensor placement and data analysis
DO-010	Operating range for lumen inner diameter within which the mechanism holds its stated behavior	UNKNOWN	details of sensor placement and data analysis
DO-011	Operating range for lumen length within which the mechanism holds its stated behavior	UNKNOWN	details of sensor placement and data analysis
DO-012	Operating range for surface roughness within which the mechanism holds its stated behavior	UNKNOWN	details of sensor placement and data analysis
DO-013	Operating range for operating pressure head within which the mechanism holds its stated behavior	UNKNOWN	details of sensor placement and data analysis
DO-014	Operating range for flow rate range within which the mechanism holds its stated behavior	UNKNOWN	details of sensor placement and data analysis
DO-015	Operating range for viscosity of flowed medium within which the mechanism holds its stated behavior	UNKNOWN	details of sensor placement and data analysis
DO-016	Sensing -> decision -> actuation loop maintaining the proposed operating point	PROPOSED	details of sensor placement and data analysis
DO-017	Information path: sensing element -> decision/monitor control/act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in	CONCEPTUAL	details of sensor placement and data analysis
DO-018	Fixture enabling: bench flow-loop characterization across the operating range — rig	UNKNOWN	details of sensor placement and data analysis
DO-019	Fixture enabling: pressure-drop mapping vs flow (identifying flow transitions) — rig	UNKNOWN	details of sensor placement and data analysis

ID	Description	Status	Missing Inputs
DO-020	Fixture enabling: accelerated occlusion / fouling challenge	UNKNOWN — rig concept finalized, build type, fabrication	UNKNOWN — scale-up, instrumentation; acceptance
DO-021	Fixture enabling: pressure-flow mapping across full operating envelope	UNKNOWN — rig concept generated, build type, fabrication	UNKNOWN — scale-up, instrumentation; acceptance

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
lumen inner diameter	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
lumen length	UNKNOWN (no sourced value)	m	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
surface roughness	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
operating pressure head	UNKNOWN (no sourced value)	Pa	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
flow rate range	UNKNOWN (no sourced value)	m^3/s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
viscosity of flowed medium	UNKNOWN (no sourced value)	Pa.s	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
occlusion growth tolerance	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
downstream occlusion	Failure of downstream occlusion	design detection and NOC: ESR and VES and DR (reduces design unknowns, triggered candidate	UNKNOWN	design candidate
proximal occlusion by tissue ingrowth	Failure of proximal occlusion by tissue ingrowth	by ESR and VES and DR (reduces design unknowns, triggered candidate	UNKNOWN	design candidate
lumen collapse under external pressure	Failure of lumen collapse under external pressure	by ESR and VES and DR (reduces design unknowns, triggered candidate	UNKNOWN	design candidate
encrustation / mineral deposition	Failure of encrustation / mineral deposition	by ESR and VES and DR (reduces design unknowns, triggered candidate	UNKNOWN	design candidate
regime transition to turbulence	Failure of regime transition to turbulence	by ESR and VES and DR (reduces design unknowns, triggered candidate	UNKNOWN	design candidate
check-valve seat wear	Failure of check-valve seat wear	by ESR and VES and DR (reduces design unknowns, triggered candidate	UNKNOWN	design candidate

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
downstream occlusion detection failure	Failure of downstream occlusion detection	designer controlled (NOT ESTABLISHED)	designer controlled (NOT ESTABLISHED)
proximal occlusion by tissue ingrowth	Failure of proximal occlusion by tissue ingrowth	material selection; patient education; follow-up	material selection; patient education; follow-up
lumen collapse under external pressure	Failure of lumen collapse under external pressure	material selection; reinforcement	material selection; reinforcement
encrustation / mineral deposition	Failure of encrustation / mineral deposition	surface coating; cleaning protocol	surface coating; cleaning protocol
regime transition to turbulence	Failure of regime transition to turbulence	altering geometry; flow conditioning	altering geometry; flow conditioning
check-valve seat wear opening	Failure of check-valve seat wear opening	material selection; replacement	material selection; replacement

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Expose the system to various occlusion scenarios	Expose the system to various occlusion scenarios	Expose the system to various occlusion scenarios	Expose the system to various occlusion scenarios
VF-002	run killer experiment: falsification test	run killer experiment: falsification test	run killer experiment: falsification test	run killer experiment: falsification test
VF-003	bench flow-loop characterization across operating range	bench flow-loop characterization across operating range	bench flow-loop characterization across operating range	bench flow-loop characterization across operating range
VF-004	pressure-drop mapping vs flow (identify pressure drops)	pressure-drop mapping vs flow (identify pressure drops)	pressure-drop mapping vs flow (identify pressure drops)	pressure-drop mapping vs flow (identify pressure drops)
VF-005	delivered-flow trend below baseline in adverse conditions	delivered-flow trend below baseline in adverse conditions	delivered-flow trend below baseline in adverse conditions	delivered-flow trend below baseline in adverse conditions
VF-006	kink/collapse test across duty cycle; flow kink/collapse test	kink/collapse test across duty cycle; flow kink/collapse test	kink/collapse test across duty cycle; flow kink/collapse test	kink/collapse test across duty cycle; flow kink/collapse test
VF-007	pressure-flow mapping across the full operating range	pressure-flow mapping across the full operating range	pressure-flow mapping across the full operating range	pressure-flow mapping across the full operating range
VF-008	reverse-leakage rate measurement after accelerated cycling	reverse-leakage rate measurement after accelerated cycling	reverse-leakage rate measurement after accelerated cycling	reverse-leakage rate measurement after accelerated cycling

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)	NOT ESTABLISHED (defined after verification)

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	medical-grade silicone elastomer	standard CSF shunt catheter material	True	ENGINEERING_PROPOSED
domain candidate material	polycarbonate / PSU rigid components	standard shunt valve housings	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
1	medical-grade silicone elastomer	1	Material	medical-grade silicone elastomer	NOT ESTABLISHED	NOT ESTABLISHED	Applicability verification required
2	polycarbonate / PSU rigid components	1	Material	polycarbonate / PSU rigid components	NOT ESTABLISHED	NOT ESTABLISHED	Applicability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- medical extrusion with inner-diameter tolerance control: domain registry manufacturing route - candidate only; no process qualification exists
- injection molding of valve/housing features: domain registry manufacturing route - candidate only; no process qualification exists
- laser drilling of calibrated orifices: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

Source 2:

URL :

Excerpt:

Source 3:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-003', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter lumen inner diameter [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter lumen length [r m]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter surface roughness [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter operating pressure head [dP Pa]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter flow rate range [Q m^3/s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter viscosity of flowed medium [mu Pa.s]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'value of critical parameter occlusion growth tolerance [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-007'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}

- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}
- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}
- {'unknown': 'acceptance criterion for verification VF-007', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-007'}
- {'unknown': 'acceptance criterion for verification VF-008', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-008'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implementing the proposed physical verification (0.5 GPa) test	Expose the system to 0.5 GPa	NOT ESTABLISHED	Article for VF-001	1000 (to be quoted)
WP-02	constraint-compliance envelope article	per method	constraint-compliance envelope article	NOT ESTABLISHED	Article for VF-002	1000 (to be quoted)
WP-03	article for VF-001	per method	Expose the system to 0.5 GPa	NOT ESTABLISHED	Article for VF-001	1000 (to be quoted)
WP-04	article for VF-002	per method	run killer experiment: 0.5 GPa	NOT ESTABLISHED	Article for VF-002	1000 (to be quoted)
WP-05	article for VF-003	per method	bench flow-loop characterization	NOT ESTABLISHED	Article for VF-003	1000 (to be quoted)
WP-06	article for VF-004	per method	pressure-drop mapping	NOT ESTABLISHED	Article for VF-004	1000 (to be quoted)
WP-07	article for VF-005	per method	delivered-flow trend	NOT ESTABLISHED	Article for VF-005	1000 (to be quoted)
WP-08	article for VF-006	per method	kink/collapse test across 0.5 GPa	NOT ESTABLISHED	Article for VF-006	1000 (to be quoted)
WP-09	article for VF-007	per method	pressure-flow mapping	NOT ESTABLISHED	Article for VF-007	1000 (to be quoted)
WP-10	article for VF-008	per method	reverse-leakage rate	NOT ESTABLISHED	Article for VF-008	1000 (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 10-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.